

YOUSSEF MOHAMED

AI Engineer | Python Backend Engineer

New Cairo, Egypt

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[Portfolio](#) | [Linkedin](#) | [Github](#)

SUMMARY

AI Engineer with hands-on experience designing and building AI-powered backend systems, intelligent automation solutions, and scalable business applications using Python. Experienced in developing production-ready applications powered by Large Language Models (LLMs), RAG pipelines, vector databases, real-time streaming, and cloud technologies. Developed NarrIQ, an AI-powered SaaS platform for automating sentiment analysis and qualitative coding, and engineered an Interactive AI Portfolio Assistant featuring low-latency voice-to-voice capabilities and semantic search. Passionate about transforming complex business challenges into scalable, production-ready AI solutions.

EDUCATION

Bachelor’s degree in Electronics & Communication Engineering | *Cairo University, Faculty of Engineering*

September 2020 – September 2025

- Grade: **Very Good**
- **Senior Project:** Advanced Automotive Radar System on TI AWR2944EVM Grade - **Excellent**

GRADUATE PROJECT

Advanced Automotive Radar System Development sponsored by *Brightskies*

September 2024 – July 2025

- Developed a 76–81 GHz radar prototype on TI AWR2944EVM, covering signal processing, antenna design, and hardware integration.
- Implemented beat-frequency estimation (FFT with optimized windowing), high-resolution MUSIC analysis, and CFAR adaptive detection.
- Deployed DBSCAN + Extended Kalman Filter for multi-target tracking
- Built CNN+GRU models for radar-based gesture recognition (> 93 % accuracy)
- Integrated real-time visualization via TI’s mmWave SDK, configured chirp profiles and managed Ethernet data acquisition.

PROFESSIONAL EXPERIENCE

- **Software & AI Solutions Engineer** | *IPSOS Egypt, Cairo*  
February 2025– Present
  - Architected and deployed DataWhisperer, a modular AI-powered internal platform featuring validation management, automated time-checking, conflict detection, RBAC, and workflow automation, significantly improving team productivity and operational efficiency.
  - Designed and developed Python automation solutions that eliminated repetitive validation tasks, significantly reducing processing time while minimizing manual effort and human error.
  - Developed scalable backend applications using Django, PostgreSQL, Celery, and Redis, enabling asynchronous processing, secure multi-user access, and reliable business operations.
  - Collaborated with operational teams to analyze business requirements and transform manual workflows into scalable software solutions that improved efficiency across multiple processes.
  - Maintained and continuously enhanced internal AI applications based on user feedback, improving usability, performance, and operational reliability.
- **Data Validation Specialist** | *IPSOS Egypt, Cairo*  
September 2023 – February 2025
  - Validated and analyzed 1,000+ survey records daily, ensuring logical consistency, data integrity, and compliance with market research quality standards.
  - Performed cross-dataset consistency checks, anomaly detection, and pattern analysis across regional market research studies.
  - Worked closely with fieldwork and operations teams to investigate data discrepancies and deliver high-quality research datasets.
  - Identified repetitive operational bottlenecks and independently developed Python automation tools, laying the foundation for internal software systems later adopted by the team.

## PROJECTS

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- **AI-Powered Portfolio Assistant | Personal Portfolio**

[Github](#) | [Portfolio](#)

- **Tech Stack:**

- Python • FastAPI • Qdrant (Vector DB) • Gemini API • RAG (Retrieval-Augmented Generation) • STT & TTS
  - Docker • Nginx • AWS EC2 • HTML/CSS/JS

Developed and deployed an interactive, production-ready AI Portfolio Assistant that acts as an autonomous agent, handling user queries about my education, experience, skills, and projects in real-time.

- Implemented an advanced RAG (Retrieval-Augmented Generation) pipeline using Qdrant Vector Database to chunk, embed, and semantically retrieve portfolio details, ensuring high contextual accuracy and eliminating LLM hallucinations.
  - Engineered a real-time, low-latency communication layer using FastAPI and Streaming Responses to deliver instantaneous, word-by-word assistant answers.
  - Integrated Voice-to-Voice capabilities by implementing Speech-to-Text (STT) and Text-to-Speech (TTS) models, enabling users to seamlessly converse with the assistant naturally via audio.
  - Containerized the entire multi-service ecosystem (API, Qdrant DB, and Nginx) using Docker Compose to guarantee consistent environment parity between local development and production.
  - Deployed the platform on AWS (EC2), configuring Nginx as a reverse proxy, securing connection traffic with SSL/TLS certificates (HTTPS) via Certbot, and routing custom domain paths smoothly.

- **NarriQ | AI-Powered Market Research Automation Platform**

[Github](#) | [Portfolio](#)

- **Tech Stack:**

- Python • Django • PostgreSQL • Celery • Redis • Docker • AWS • OpenAI API • Embeddings • Semantic Clustering

Developed an AI-powered SaaS platform that automates open-ended coding, sentiment analysis, and qualitative text processing for market research workflows.

- Designed and implemented end-to-end LLM pipelines using OpenAI APIs, semantic embeddings, and clustering techniques to automate qualitative text analysis with over 90% sentiment classification accuracy.
  - Engineered intelligent response-grouping algorithms that significantly reduced LLM API calls, lowering inference costs while maintaining high classification quality.
  - Developed scalable asynchronous processing using Celery and Redis, enabling concurrent analysis jobs across multiple users and projects.
  - Built secure multi-tenant backend architecture with RBAC, authentication, subscription management, and PostgreSQL for production-ready deployment.
  - Containerized and deployed the platform using Docker and AWS, providing a scalable and production-ready cloud environment.
  - Reduced manual qualitative coding workflows from days to hours, significantly improving operational efficiency for market research teams.

- **DataWhisperer | Internal Validation Management Platform**

[Github](#) | [Portfolio](#)

- **Tech Stack:**

- Python • Django • PostgreSQL • Celery • Redis • Nginx • JavaScript • HTML • CSS

Designed and developed an internal operations platform that centralized validation management, interviewer analytics, workflow automation, and operational reporting for market research teams.

- Designed and developed a Validation Management System that maintains validation history across multiple projects, tracks validation rounds, and monitors issue resolution.
  - Engineered an automated Time Check engine capable of detecting interview conflicts, out-of-working-hours interviews, interview length (LOI) anomalies, and daily interview activity.
  - Designed secure authentication and role-based access control (RBAC) supporting multiple user roles and controlled access to operational modules.
  - Developed interactive dashboards providing real-time project progress, validation metrics, interviewer performance, operational KPIs, and historical trend analytics to support data-driven decision-making.
  - Implemented detailed interviewer performance profiles featuring yearly error-rate trends, project participation history, interview statistics, and issue breakdown analytics.
  - Developed Excel import/export pipelines for processing validation datasets and generating automated operational reports.
  - Integrated asynchronous task processing using Celery and Redis to efficiently execute large-scale validation and reporting workflows.
  - Deployed and maintained the platform on the company's internal infrastructure, ensuring secure and reliable daily operations for validation teams.

TECHNICAL SKILLS

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- **Programming Languages:** Python, JavaScript, SQL, C/C++.
- **Backend & Web:** Django, FastAPI, REST APIs, Celery, Redis, Nginx, HTML5, CSS3.
- **Databases:** PostgreSQL.
- **AI/LLM:** OpenAI API, Gemini API, Prompt Engineering, Embeddings, Semantic Search, Semantic Clustering, NLP Pipelines, RAG (Retrieval-Augmented Generation), Qdrant (Vector DB).
- **Cloud & DevOps:** Docker, AWS, Git, Linux.
- **Tools:** MATLAB.
- **Core Concepts:** Object-Oriented Programming (OOP), Data Structures, Algorithms, Asynchronous Processing, Authentication & RBAC.

LANGUAGES

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- **Arabic:** Native
- **English:** Professional